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UNIVERSITY OF WOLVERHAMPTON

HERALD COLLEGE KATHMANDU

Big Data (6CS030)

Workshop-2 –
Report on Subject Specific Big Data Process

Tutor : Ms. Rakhee Pandey

Group : L6CG10

Members : Aryan Lamichhane, Rujan Maharjan, Utsav Jarga Magar

Introduction

This report outlines a comprehensive big data processing workflow on the “Spotify Tracks: Audio Features (50k Songs)’ dataset sourced from Kaggle which consists of the data of 50,000 synthetic Spotify-style tracks spanning form 2000 - 2024 across 20 genres. The dataset mirrors the Spotify Audio Features API that consist of: danceability, energy, loudness, speechiness, accousticness, instumentalness, liveness, valence, tempo, duration, key and mode all generated with realistic per-genre distribution and correlations for example like electronic tracks having high energy plus danceability. By analyzing these features our objective is to identify the top ten genres across various timeframes based on popular metrics. Through data ingestion, transformation and visualization we aim to provide a way to understand the listener’s preferences through analyzing audio metadata.

The primary data source for this project is the “Spotify Tracks: Audio Features (50k Songs)’ dataset which is stored in a structured CSV format. To ensure high availability and seamless integration with our processing environment, we uploaded the dataset to google drive. This allows our compute engine (Google Colab) to access the 50,000-record dataset without local hardware constraints. To automate the data acquisition process and ensure the workflow is reproducible, we utilize the google collab library to programmatically mount the drive. This eliminates the need for manual file uploads during each session.

Data Exploration

Data Cleaning

```python
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

```python
dataset = pd.read_csv('/content/drive/MyDrive/Big Data/Workshop2/spotify_tracks.csv')
dataset.info()
dataset.shape
dataset.head()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   track_id          50000 non-null  object
 1   track_name        50000 non-null  object
 2   artist_name       50000 non-null  object
 3   album_name        50000 non-null  object
 4   release_year      50000 non-null  int64
 5   genre             50000 non-null  object
 6   popularity        50000 non-null  int64
```

To begin the Data Exploration phase, we used the .info() and .head() methods to validate the structural integrity of the 50,000-track dataset. This allowed us to confirm that all audio features (e.g., danceability, valence) were correctly loaded as numerical values, which is essential for the subsequent popularity-based ranking and trend analysis

```
else:
  print("Duplicate rows has been deleted")
```

```
track_id            0
track_name          0
artist_name         0
album_name          0
release_year        0
genre               0
popularity          0
duration_ms         0
explicit            0
danceability        0
energy              0
loudness            0
speechiness         0
acousticness        0
instrumentalness    0
liveness            0
valence             0
tempo               0
key                 0
mode                0
time_signature      0
dtype: int64
/n
50000
50000


There are no duplicate rows
```

To ensure the accuracy of our genre popularity analysis, we performed a two-stage data cleaning process. First, we conducted a **Null Value Audit**. Identifying missing values early is critical to prevent bias in audio feature correlations.

In large-scale datasets, duplicate entries can lead to 'over-weighting' certain tracks, which would distort our 'Top 10 Genres' findings. By comparing the dataset length before and after the drop duplicate function, we verified the uniqueness of our 50,000 tracks. This step ensures that our insights are based on a diverse and non-redundant sample of audio metadata.

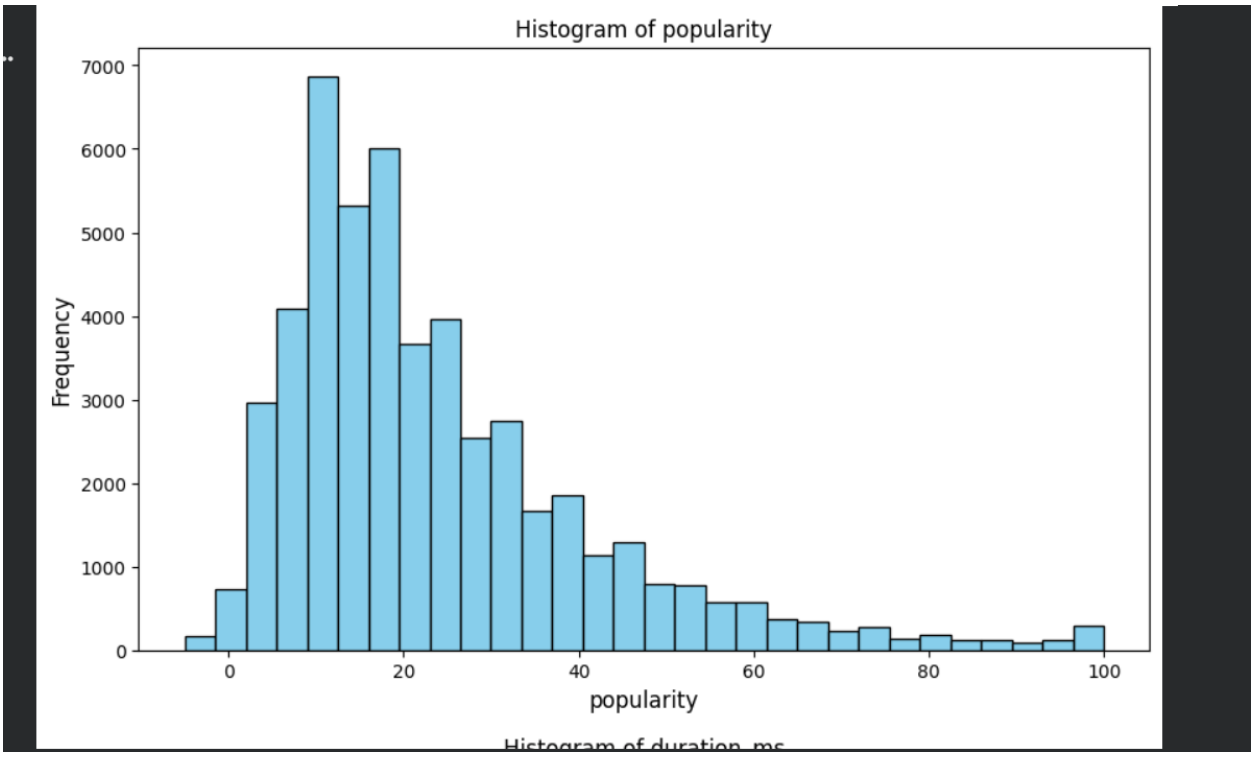
```
dataset.describe()
```

	release_year	popularity	duration_ms	danceability	energy	loudness	speechiness	acousticness	instrumentalness	liveness	valence	tempo
count	50000.000000	50000.000000	50000.000000	50000.000000	50000.000000	50000.000000	50000.000000	50000.000000	50000.000000	50000.000000	50000.000000	50000.000000
mean	2015.853400	23.599560	209976.955080	0.626921	0.658879	-7.438755	0.076370	0.275942	0.155218	0.181120	0.581443	116.258270
std	6.178209	17.912623	44881.858907	0.207513	0.212950	4.305645	0.064989	0.257911	0.235443	0.096788	0.183994	23.706129
min	2000.000000	-5.000000	90000.000000	0.000000	0.000000	-27.795904	0.000000	0.000000	0.000000	0.000000	0.000000	50.000000
25%	2012.000000	11.000000	179812.250000	0.489439	0.539592	-9.802727	0.031339	0.075444	0.000000	0.112240	0.457651	100.238179
50%	2017.000000	19.000000	209886.500000	0.651486	0.683479	-6.906656	0.062609	0.198993	0.064522	0.179683	0.581160	115.325911
75%	2021.000000	31.000000	240023.750000	0.783254	0.812390	-4.425697	0.098898	0.412742	0.168282	0.247669	0.708097	130.455924
max	2024.000000	100.000000	403583.000000	1.000000	1.000000	0.000000	0.368468	1.000000	1.000000	0.581942	1.000000	220.000000

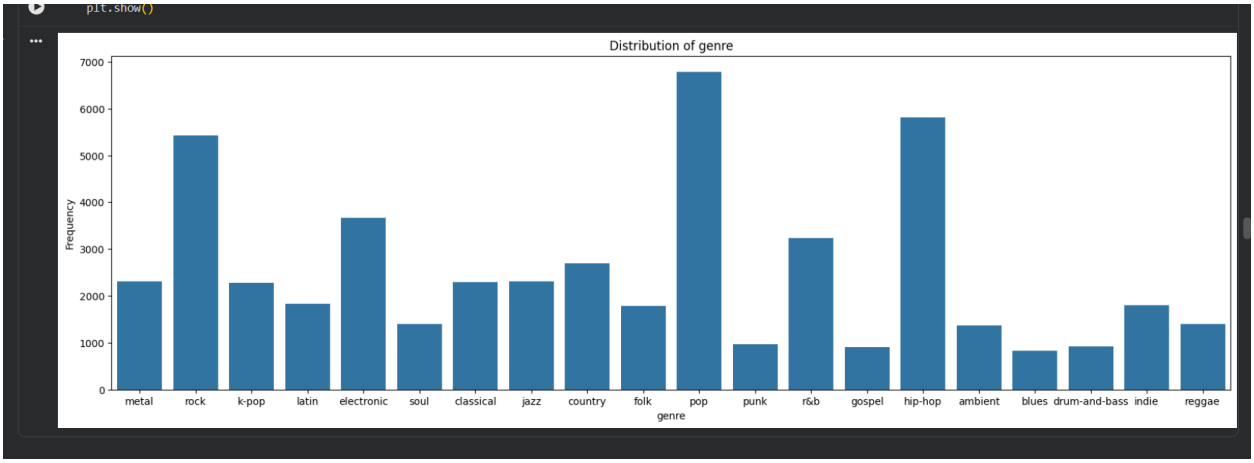
.

```
Mode values for features:
release_year          2023.0
popularity              13.0
duration_ms          90000.0
danceability             1.0
energy                   1.0
loudness                 0.0
speechiness              0.0
acousticness             0.0
instrumentalness         0.0
liveness                 0.0
valence                  1.0
tempo                   50.0
key                     11.0
mode                     1.0
time_signature           4.0
dtype: float64
```

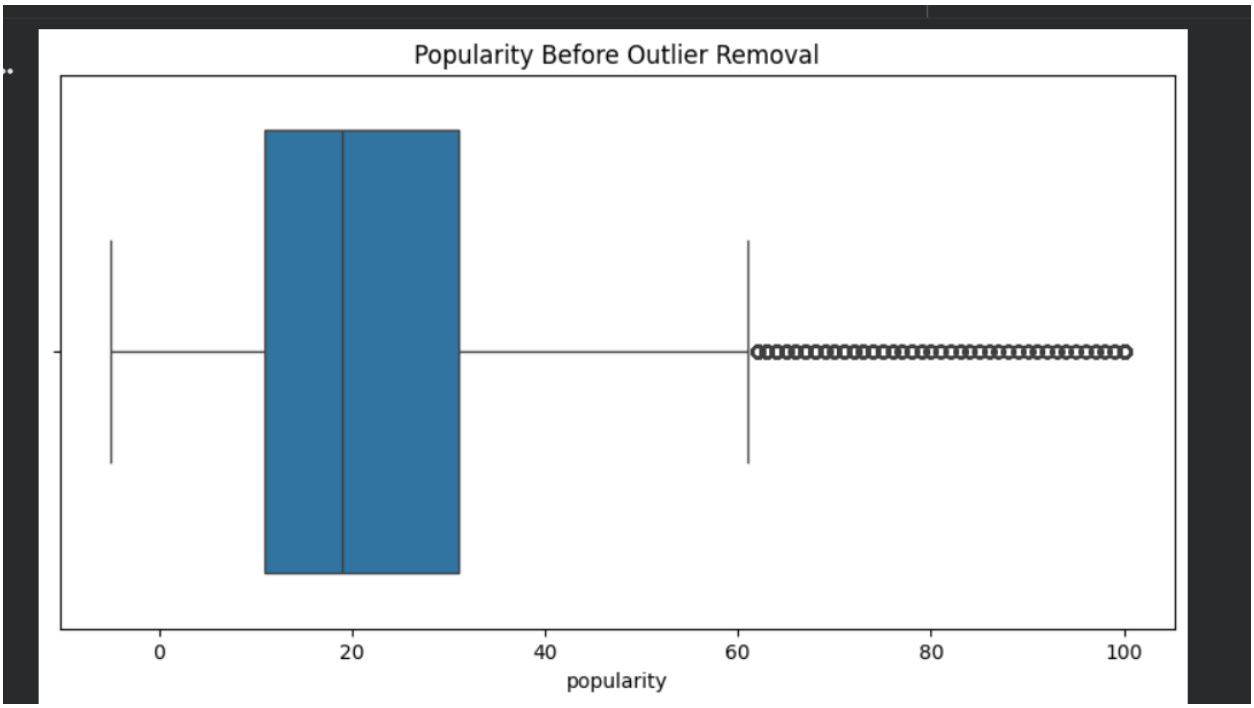
We used .describe function to find the mean, median, Quartile, standard deviation, etc. of each numerical rows

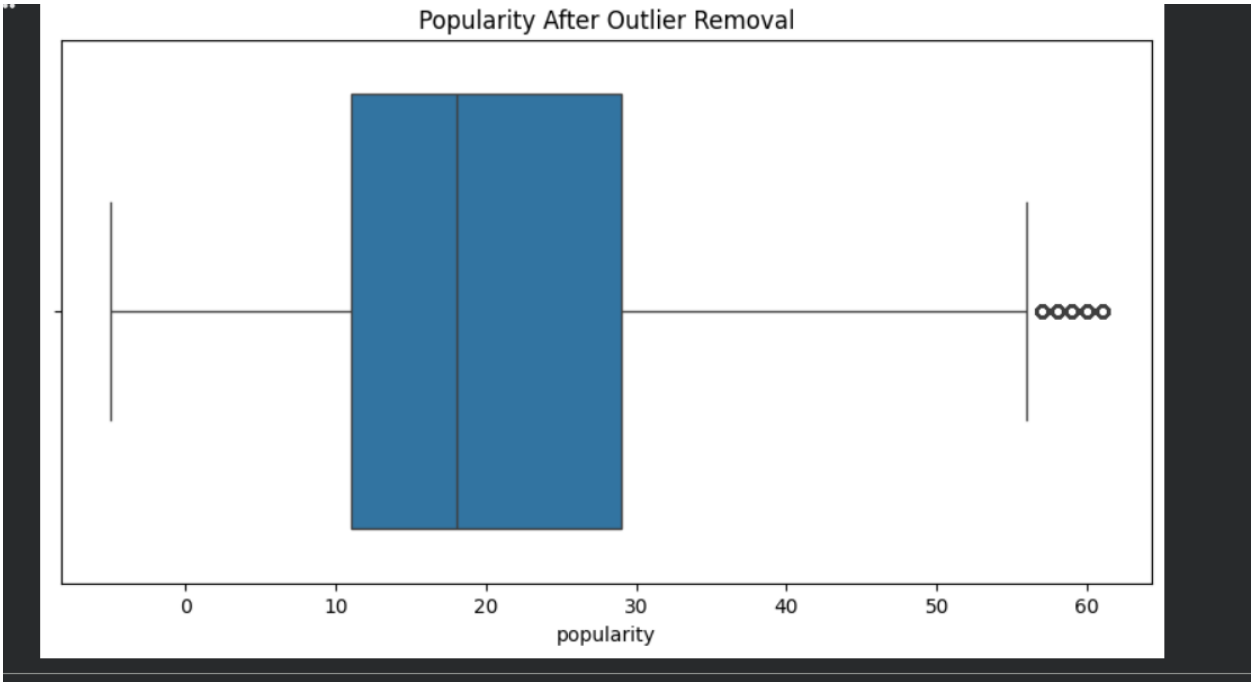


This here is the Hitogram of Popularity column. It shows how many datapoints fall in each range called bins. If popularity between ranges from 0 to 100 and most songs are between 0 to 20.

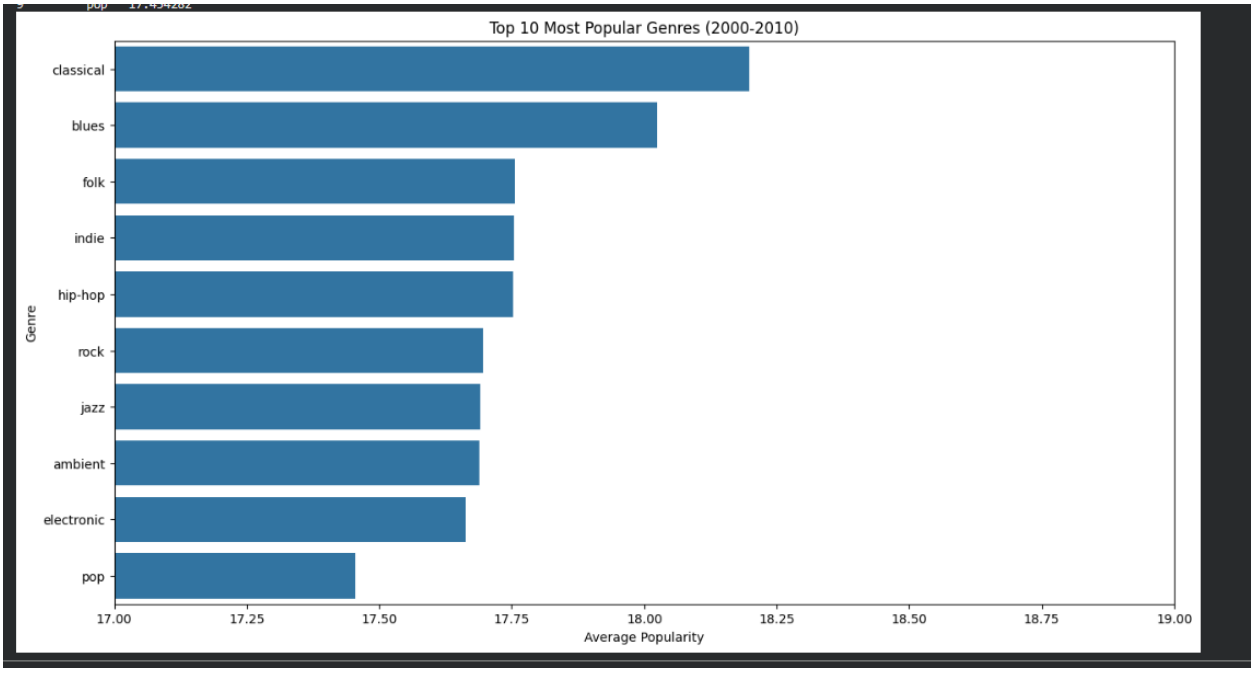


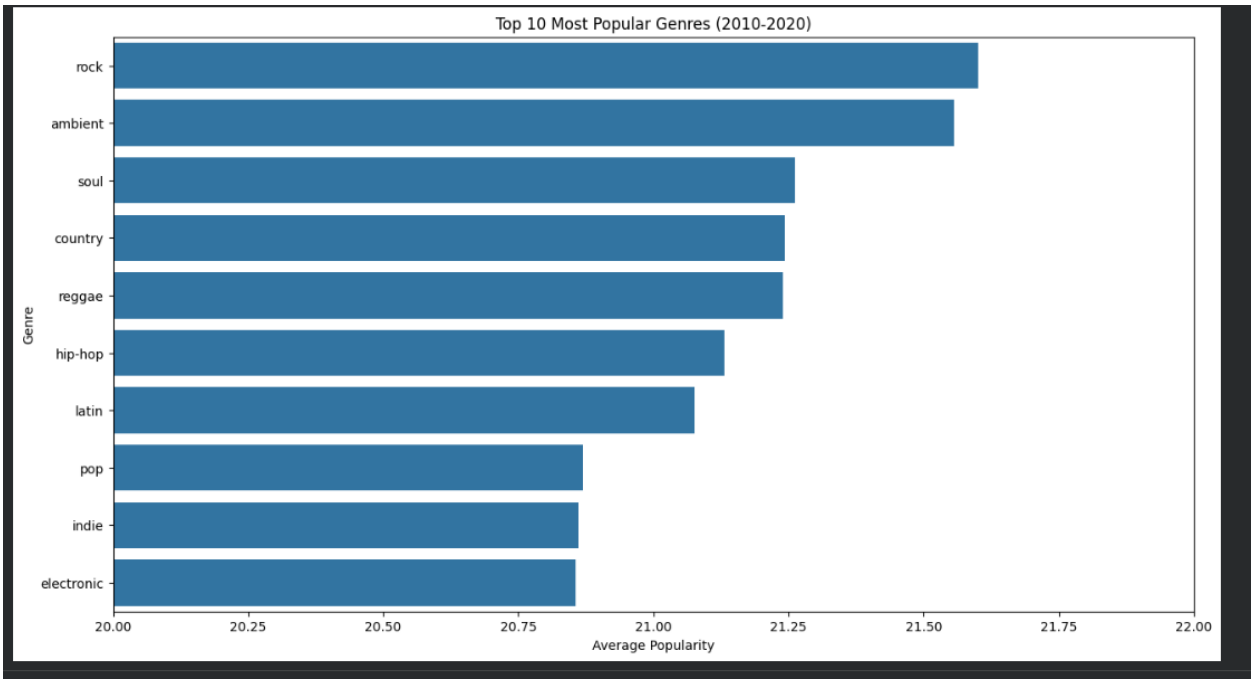
This here is the distribution of the genre across the metrics shown in a bar graph.

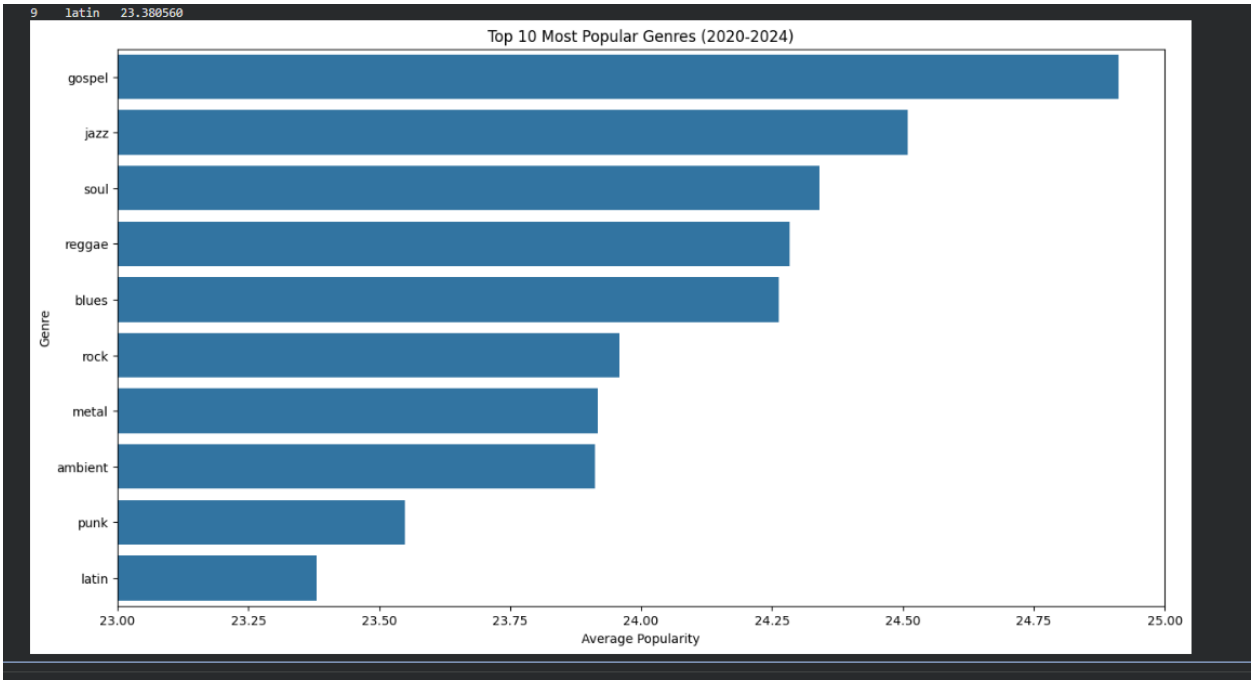




This are the before and after handleing the outliers in popularity.


Top 10 Most Popular Genres (2000-2010)
classical
blues
folk
indie
hip-hop
rock
jazz
ambient
electronic
pop
Genre
17.00
17.25
17.50
17.75
18.00
18.25
18.50
18.75
19.00
Average Popularity


Top 10 Most Popular Genres (2010-2020)
rock
ambient
soul
country
reggae
hip-hop
latin
pop
indie
electronic
Genre
20.00
20.25
20.50
20.75
21.00
21.25
21.50
21.75
22.00
Average Popularity



Here are the final distribution of the genres according to popularity across every 10 years.

Here is the co-relation between all the numerical values in the dataset.

Github Link:

https://github.com/TriBot2o/BigData_Year3